

Aaron Dutta

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EDUCATION

University of Waterloo

Expected June 2031

BMath, Honours Mathematics with Co-Op

Waterloo, ON

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML5/CSS3, SQL

Libraries/Frameworks: FastAPI, Flask, Next.js, Prisma, React, Streamlit, Tailwind CSS

Developer Tools: Anthropic API, Docker, Figma, Gemini API, Git, MediaPipe, Microsoft Azure, NetworkX, NumPy, ONNX Runtime, OpenCV, Pandas, PostgreSQL, Pydantic, RESTful APIs, SQLite, Supabase, Vercel, YOLOv8

PROJECTS

TechniqueTitan | *Python, MediaPipe, OpenCV, NumPy, FastAPI, Pandas, Matplotlib* |   2026

- Built an **AI piano posture coach** that turns any device camera into **on-demand technique feedback** so students can correct hand form without a teacher
- Engineered a **MediaPipe + OpenCV + NumPy** pipeline to extract **21 landmarks** per hand, normalize for camera distance, and score **5 posture criteria**
- Shipped a **FastAPI + React/TypeScript** app with on-device browser MediaPipe and YAML-tunable thresholds, cutting review from a full lesson cycle to seconds
- Built a **batch CLI** that walks image folders and exports **CSV/JSON** metrics along with a 45-test pytest suite across geometry, scoring, and API modules, automating dataset-scale analysis that was previously manual

SkyEye | *Python, Flask, YOLOv8, ONNX Runtime, React, TypeScript, Gemini API* |   2026

- Won **RescueHacks 2026** for an SAR triage tool that turns a free-text missing-person report into a **Lost Person Behavior** search ring, scoring map pins in **5–8 s** within a **30 s** time-to-first-lead target
- Developed a **sliding-window YOLOv8n + ONNX Runtime** pipeline to tile images into overlapping windows
- Shipped a **Flask + React/TypeScript** operator console with **Gemini Flash** (Groq fallback) for typed extraction and **Google Geocoding + LPB $\sqrt{t}$ radius**

FraudGen | *Python, Anthropic API, FastAPI, Streamlit, Pandas, NetworkX, Pydantic* |   2026

- Designed a **4-agent adversarial LLM** pipeline that explores fraud variant space via a **coverage matrix** and generates **persona-seeded mule-network** scenarios
- Shipped **async parallel orchestration**, deterministic **8-rule** validation/repair, and **critic scoring** with retry loops; a single produced variant labeled **11 transactions** across a 3-victim $\rightarrow$ 4-mule converging funnel
- Built a **Streamlit** console with **network graph** visualization and **ML-ready exports** for **GNN** training

CERTIFICATIONS

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|---|------|
| Essentials with Azure Fundamentals, Microsoft | 2026 |
| Google AI Essentials Specialization, Google | 2026 |
| Building with the Claude API, Anthropic | 2026 |
| CS50x, Harvard University | 2024 |

AWARDS

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|---|------|
| Winner @ RescueHacks | 2026 |
| 2nd Place Overall @ Katy Youth Hacks | 2026 |
| 3rd Place Security/Privacy Hack @ STEMist Hacks | 2026 |
| 1st Place @ STEMMINDS Hackathon | 2026 |

EXPERIENCE

Mathematics Mentor

2024 – Present

Filix School of Education

Purulia, India (Remote)

- Mentoring **15+** Grade 7–12 students through weekly math circles, encouraging collaborative problem solving
- Advancing students' critical thinking skills by combining theoretical concepts with practical exercises